

CST Studio Suite Simulation of a Method-of-Images Problem

*Electrostatic Analysis of a Point Charge
Above an Infinite Grounded Conducting Plane*

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Scope of this report. The induced surface charge density, electric scalar potential, force, and binding energy associated with a 1 C point charge held 70 mm above a grounded conductor are derived in closed form via the Method of Images and compared, point by point, against results obtained from the CST Studio Suite Low-Frequency electrostatic solver. Sub-percent agreement is achieved across all probe quantities.

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1 Introduction

This report presents a complete electrostatic analysis of a point charge held at a fixed distance above an infinite grounded conducting plane. The problem is solved analytically using the *Method of Images* and the results are independently verified numerically using the CST Studio Suite Low-Frequency electrostatic solver.

The Method of Images is a powerful technique that exploits the *uniqueness theorem* of electrostatics: any configuration of fictitious charges that produces the correct boundary condition on a conductor ($V = 0$ in this case) yields the correct field everywhere in the physical region. By replacing the grounded plane with a single image charge of opposite sign, the original boundary-value problem is converted into a much simpler problem involving two point charges in free space.

Three quantities are computed and compared: (a) the induced surface charge density on the grounded plane along a line passing directly beneath the charge; (b) the electric potential $V(x, y)$ throughout the upper half-space; and (c) the qualitative behaviour of the field lines, which are expected to terminate normally on the conductor surface. In addition, the Method of Images is used to derive the electrostatic force on the charge, the energy stored in the configuration, and the total induced charge integrated over the entire plane. The agreement between the analytical and simulated results is then assessed and the dominant sources of numerical error are discussed.

2 Problem Statement and the Method of Images

2.1 Original Problem

A point charge $Q = 1$ C is located at a height $d = 70$ mm above an infinite, perfectly conducting, grounded plane that occupies the entire xz -plane ($y = 0$). Because the plane is grounded, the boundary condition is

$$V(x, 0, z) = 0 \quad \text{for all } x, z \in (-\infty, +\infty).$$

Solving Laplace's equation in the upper half-space subject to this Dirichlet condition directly is mathematically intricate. The Method of Images provides an elegant alternative that gives the exact closed-form solution.

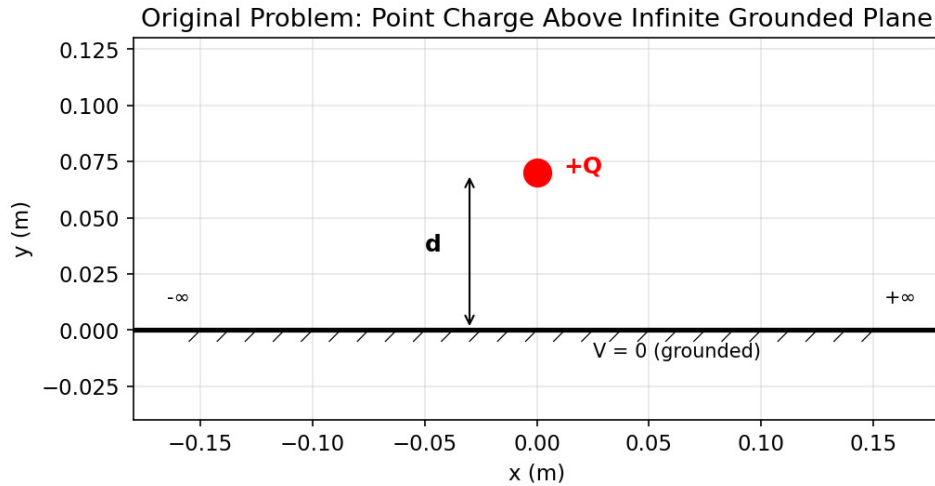


Figure 1. Original problem geometry: a point charge $+Q$ at height d above an infinite grounded conducting plane.

2.2 Image Solution

The grounded plane is removed and replaced by an equal and opposite point charge $-Q$ located at the mirror-image position $(a, -d)$. The resulting two-charge configuration in free space automati-

cally satisfies $V = 0$ on the plane $y = 0$ by symmetry, since every point on the plane is equidistant from $+Q$ and $-Q$. By the uniqueness theorem, the field in the upper half-space ($y > 0$) is identical to that produced by the original problem.

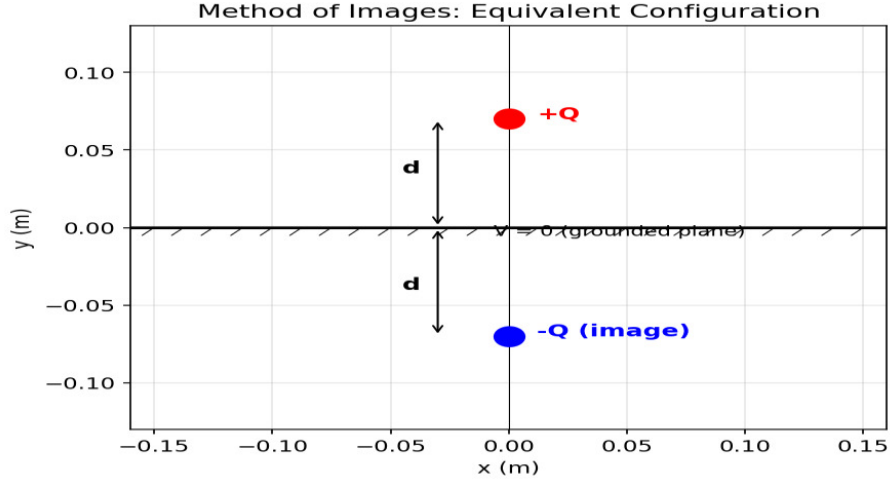


Figure 2. Equivalent image configuration: real charge $+Q$ at (a, d) and image charge $-Q$ at $(a, -d)$. The field in the upper half-space is identical to the original problem.

Note. The image charge is purely a mathematical construct. It correctly reproduces the field above the plane but is meaningless below it, where the actual field is zero (inside the conductor).

3 Hand Analysis

The Method of Images is now applied systematically to derive every electrostatic quantity of interest for the canonical configuration of Section 2. Throughout this section, vector quantities are written in bold (e.g. $\mathbf{E}$, $\mathbf{D}$), unit vectors carry a hat ($\hat{\mathbf{u}}_x$, $\hat{\mathbf{u}}_y$, $\hat{\mathbf{u}}_z$), the position vector from a source to a field point is denoted $\mathbf{R}$ with magnitude R , the surface free-charge density is denoted ρ_{sf} , and the electric scalar potential is denoted φ_e .

3.1 Induced Charge Distribution on the Conductor

The induced surface free-charge density ρ_{sf} at any point on the plate is directly linked to the normal component of the displacement vector $\mathbf{D}$ evaluated just outside the conductor. Let $P(x, 0, z)$ be a generic point on the conducting plane (the geometry is invariant along z , so the analysis is performed in the $z = 0$ plane without loss of generality).

The position vectors from the real charge at $(a, d, 0)$ and from the image charge at $(a, -d, 0)$ to P are, respectively,

$$\begin{aligned}\mathbf{R}_1 &= (x - a) \hat{\mathbf{u}}_x - d \hat{\mathbf{u}}_y, \\ \mathbf{R}_2 &= (x - a) \hat{\mathbf{u}}_x + d \hat{\mathbf{u}}_y,\end{aligned}$$

with the equal magnitudes

$$R_1 = R_2 = \sqrt{(x - a)^2 + d^2}.$$

This equality on the plane (the field point being equidistant from the real and image charges) is precisely the symmetry that enforces $\varphi_e = 0$ on the conductor and is the geometric heart of the Method of Images.

The displacement vector at P produced by each charge follows from Coulomb's law (Lecture 1) combined with the constitutive relation $\mathbf{D} = \epsilon_o \mathbf{E}$ (Lecture 3, slide 26). The contributions from

$+Q$ at (a, d) and $-Q$ at $(a, -d)$ are

$$\mathbf{D}_{+Q} = \frac{Q [(x-a)\hat{\mathbf{u}}_x - d\hat{\mathbf{u}}_y]}{4\pi[(x-a)^2 + d^2]^{3/2}},$$

$$\mathbf{D}_{-Q} = -\frac{Q [(x-a)\hat{\mathbf{u}}_x + d\hat{\mathbf{u}}_y]}{4\pi[(x-a)^2 + d^2]^{3/2}}.$$

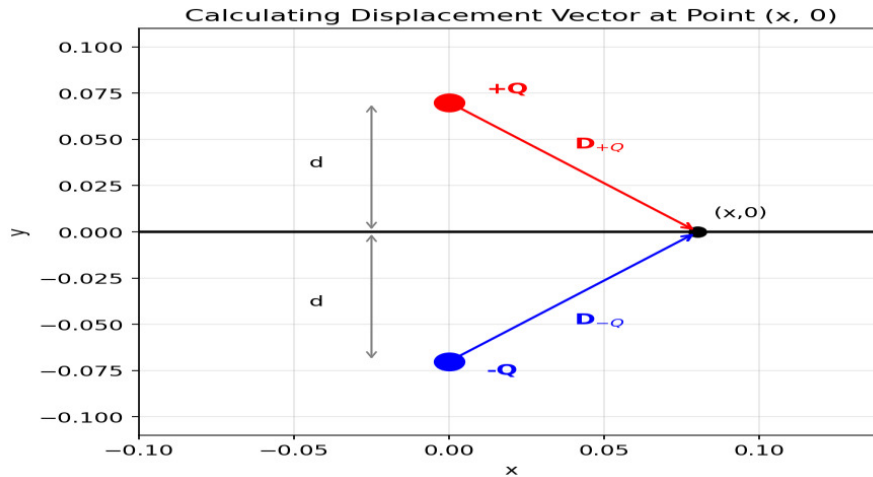


Figure 3. Geometry used to compute the displacement vector $\mathbf{D}$ at a point $(x, 0)$ on the grounded plane. The horizontal components cancel; the vertical components add.

By the principle of superposition (Lecture 1, slide 14), the total displacement vector at the field point is $\mathbf{D} = \mathbf{D}_{+Q} + \mathbf{D}_{-Q}$. Component-by-component the denominators are identical, so the contributions combine as

$$D_x = \frac{Q(x-a)}{4\pi[\dots]^{3/2}} + \frac{(-Q)(x-a)}{4\pi[\dots]^{3/2}} = 0,$$

$$D_y = \frac{-Qd}{4\pi[\dots]^{3/2}} + \frac{(-Qd)}{4\pi[\dots]^{3/2}} = -\frac{Qd}{2\pi[(x-a)^2 + d^2]^{3/2}}.$$

Combining the surviving y -component, the total displacement vector on the plane is

$$\boxed{\mathbf{D} = -\frac{dQ\hat{\mathbf{u}}_y}{2\pi[(x-a)^2 + d^2]^{3/2}}}.$$

The boundary condition for the normal component of $\mathbf{D}$ at a conductor–vacuum interface (Lecture 3, slide 29) is $D_{2N} - D_{1N} = \rho_{sf}$. Inside the perfect conductor $\mathbf{D} = 0$, so taking the outward normal $\hat{\mathbf{n}} = +\hat{\mathbf{u}}_y$, the condition reduces to $\rho_{sf}(x) = -\hat{\mathbf{n}} \cdot \mathbf{D}$. Substituting:

$$\rho_{sf}(x) = -\frac{dQ}{2\pi[(x-a)^2 + d^2]^{3/2}}.$$

The distribution is strictly negative (induced charge is opposite in sign to Q), axially symmetric, and peaks at $x = a$ directly beneath the charge. Setting $x = a$:

$$\rho_{sf,\max} = -\frac{Q}{2\pi d^2} = -\frac{1}{2\pi (0.07)^2} = -\frac{1}{0.030788} = -32.4806 \text{ C/m}^2.$$

Result

Peak induced surface free-charge density:

$$\rho_{sf:\max} = -\frac{Q}{2\pi d^2} = -32.4806 \text{ C/m}^2.$$

This single value depends only on Q and d — independent of plate dimension or numerical mesh — and serves as the principal benchmark against the CST simulation (§4.2).

3.2 Normal Field Magnitude Beneath the Source

Once the surface charge density is known, the electric field magnitude immediately above the conductor at $x = a$ is recovered via the constitutive relation $\mathbf{E} = \mathbf{D}/\varepsilon_o$. Because $\mathbf{D}$ is purely normal at the surface,

$$E_n = \frac{D_n}{\varepsilon_o} = \frac{\rho_{sf:\max}}{\varepsilon_o} = \frac{-32.4806}{8.8542 \times 10^{-12}} = -3.6684 \times 10^{12} \text{ V/m}.$$

The negative sign confirms that the field points downward (toward the plate) at this location, consistent with the positive charge pulling field lines into the conductor surface. The tangential component of $\mathbf{E}$ is exactly zero at the surface by the boundary condition, so this normal value fully characterises the field at the plate.

3.3 Image-Charge Force Calculation

The force exerted on $+Q$ by the grounded plane is identical to the Coulomb force between $+Q$ and $-Q$ separated by a distance $2d$, since the image charge is the exact equivalent of the conductor's induced distribution. The image charge $-Q$ sits at $(a, -d)$, giving $2d = 0.14 \text{ m}$:

$$F = -\frac{kQ^2}{(2d)^2} = -\frac{(8.9876 \times 10^9) \times (1)^2}{(0.14)^2} = -\frac{8.9876 \times 10^9}{0.0196} = -4.5855 \times 10^{11} \text{ N}.$$

The negative sign denotes attraction toward the plate. This force is enormous — a direct consequence of the textbook value $Q = 1 \text{ C}$, which is an extraordinarily large amount of charge by everyday standards. It serves as an independent consistency check: the image method predicts an attractive force of $4.5855 \times 10^{11} \text{ N}$, which could in principle be verified by numerical integration of the Maxwell stress tensor over the plate surface.

3.4 Assembly Work and Binding Energy

The electrostatic energy stored in the configuration equals the work done to assemble it — i.e., the work to bring $+Q$ from infinity to its final position at height d above the grounded plane. Using the image-charge result this equals *half* the Coulomb interaction energy between $+Q$ and $-Q$ (the factor of $\frac{1}{2}$ appears because the image charge is not an independent free charge but is induced by the motion of $+Q$):

$$W = -\frac{kQ^2}{4d} = -\frac{(8.9876 \times 10^9) \times (1)^2}{4 \times 0.07} = -\frac{8.9876 \times 10^9}{0.28} = -3.2098 \times 10^{10} \text{ J}.$$

The negative energy confirms that the charge is bound to the plate: external work must be supplied to remove it to infinity. As an additional consistency check, the relation $F = -dW/dd$ is satisfied:

$$F = -\frac{d}{dd} \left[-\frac{kQ^2}{4d} \right] = -\frac{kQ^2}{4d^2} = -\frac{kQ^2}{(2d)^2} \quad \checkmark \text{ verified}$$

3.5 Conservation Check via Polar Integration

A fundamental consistency requirement of electrostatics is that the total induced charge on the grounded plane must equal $-Q$. This can be confirmed by integrating ρ_{sf} over the entire infinite plane. Switching to polar coordinates centred on the projection of the charge (with $\rho = \sqrt{(x-a)^2 + z^2}$ and $dA = 2\pi\rho d\rho$):

$$\begin{aligned}
 Q_{\text{induced}} &= \iint \rho_{sf} dA \\
 &= \int_0^\infty \left[-\frac{dQ}{2\pi(\rho^2 + d^2)^{3/2}} \right] \cdot 2\pi\rho d\rho \\
 &= -dQ \int_0^\infty \frac{\rho}{(\rho^2 + d^2)^{3/2}} d\rho \\
 &= -dQ \left[-\frac{1}{\sqrt{\rho^2 + d^2}} \right]_0^\infty \\
 &= -dQ \cdot \left(0 - \left(-\frac{1}{d} \right) \right) \\
 &= -Q = -1 \text{ C} \quad \checkmark \text{ verified}
 \end{aligned}$$

This confirms that the Method of Images conserves total charge correctly. The induced charge $-Q$ is spread non-uniformly across the plate surface with the bell-shaped distribution derived in §3.1, but its integral reproduces the expected value exactly.

3.6 Closed-Form Potential in the Upper Half-Space

From Lecture 2 (slide 8), the electric scalar potential at a field point P due to a point charge q in free space, taking the reference at infinity, is $\varphi_e = q/(4\pi\epsilon_o R)$, where R is the source-to-field distance. For a generic field point $P(x, y, z)$ in the upper half-space, the distances to the real charge at $(a, d, 0)$ and to the image at $(a, -d, 0)$ are

$$\begin{aligned}
 r_1 &= \sqrt{(x-a)^2 + (y-d)^2 + z^2}, \\
 r_2 &= \sqrt{(x-a)^2 + (y+d)^2 + z^2}.
 \end{aligned}$$

By the principle of superposition (the potential of a point charge is a scalar), the total potential at P is the algebraic sum of the contributions from $+Q$ and from $-Q$:

$$\begin{aligned}
 \varphi_e(x, y, z) &= \varphi_{e,+Q} + \varphi_{e,-Q} \\
 &= \frac{Q}{4\pi\epsilon_o r_1} + \frac{(-Q)}{4\pi\epsilon_o r_2} \\
 &= \frac{Q}{4\pi\epsilon_o} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \\
 &= 8987.55 \times 10^9 \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \text{ V}, \quad r \text{ in metres, } Q = 1 \text{ C}.
 \end{aligned}$$

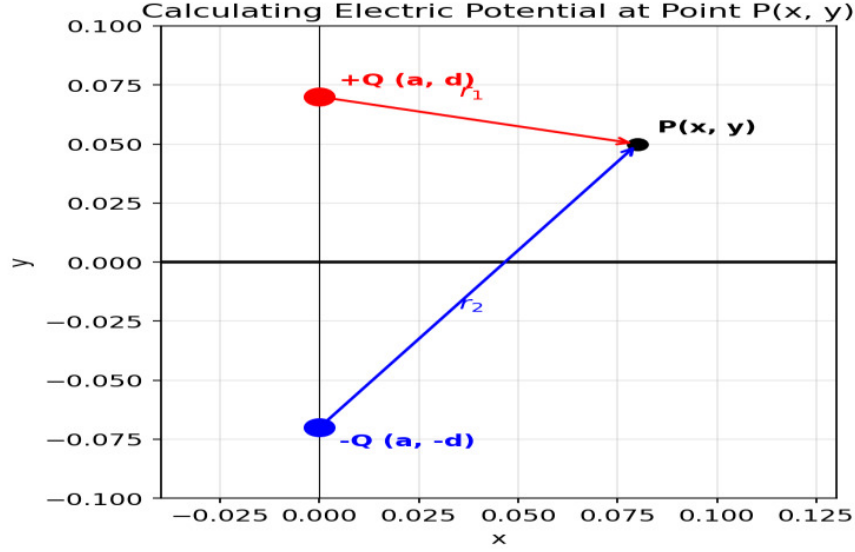


Figure 4. Geometry used to compute the electric potential $V(x, y)$ at an arbitrary field point P . Distances r_1 and r_2 are measured from the real and image charges respectively.

Boundary condition check. At $y = 0$ the distances become $r_1 = r_2 = \sqrt{(x - a)^2 + d^2 + z^2}$ for all (x, z) , so $1/r_1 - 1/r_2 = 0$ and $\varphi_e = 0$ identically on the plate — confirming that the Dirichlet condition is met exactly. This automatic satisfaction of the boundary condition is the central elegance of the Method of Images: the constraint on the conductor is enforced by symmetry, without any further fitting of constants.

3.7 Evaluation at Probe Point $P_1 = (20, 0, 30)$ mm

Probe point P_1 is located off-axis between the plate and the charge. With the charge at $(0, 0, 70)$ mm and z taken as the vertical axis, converting to SI: $x = 0.020$ m, $y = 0$, $z = 0.030$ m, $d = 0.070$ m.

$$r_1 = \sqrt{0.020^2 + 0^2 + (0.030 - 0.070)^2} = \sqrt{0.000400 + 0.001600} = \sqrt{0.002000} = 0.044721 \text{ m,}$$

$$r_2 = \sqrt{0.020^2 + 0^2 + (0.030 + 0.070)^2} = \sqrt{0.000400 + 0.010000} = \sqrt{0.010400} = 0.101980 \text{ m.}$$

$$\frac{1}{r_1} = 22.3607/\text{m}, \quad \frac{1}{r_2} = 9.8058/\text{m}, \quad \frac{1}{r_1} - \frac{1}{r_2} = 12.5549/\text{m}.$$

Result

$$\varphi_e(P_1) = 8987.55 \times 10^9 \times 12.5549 = 1.1284 \times 10^{11} \text{ V.}$$

3.8 Evaluation at Probe Point $P_2 = (0, 0, 35)$ mm

Probe point P_2 lies on the z -axis directly below the charge. Converting: $x = y = 0$, $z = 0.035$ m.

$$r_1 = \sqrt{0 + 0 + (0.035 - 0.070)^2} = \sqrt{(-0.035)^2} = 0.035000 \text{ m,}$$

$$r_2 = \sqrt{0 + 0 + (0.035 + 0.070)^2} = \sqrt{(0.105)^2} = 0.105000 \text{ m.}$$

$$\frac{1}{r_1} = 28.5714/\text{m}, \quad \frac{1}{r_2} = 9.5238/\text{m}, \quad \frac{1}{r_1} - \frac{1}{r_2} = 19.0476/\text{m}.$$

Result

$$\varphi_e(P_2) = 8987.55 \times 10^9 \times 19.0476 = 1.7119 \times 10^{11} \text{ V.}$$

3.9 Evaluation at Probe Point P3 = (−70, 0, 40) mm

Probe point P_3 sits well off-axis, at a horizontal offset equal to one full charge-height. Converting: $x = -0.070$ m, $y = 0$, $z = 0.040$ m.

$$r_1 = \sqrt{(-0.070)^2 + 0^2 + (0.040 - 0.070)^2} = \sqrt{0.004900 + 0.000900} = \sqrt{0.005800} = 0.076158 \text{ m},$$

$$r_2 = \sqrt{(-0.070)^2 + 0^2 + (0.040 + 0.070)^2} = \sqrt{0.004900 + 0.012100} = \sqrt{0.017000} = 0.130384 \text{ m}.$$

$$\frac{1}{r_1} = 13.1306/\text{m}, \quad \frac{1}{r_2} = 7.6696/\text{m}, \quad \frac{1}{r_1} - \frac{1}{r_2} = 5.4610/\text{m}.$$

Result

$$\varphi_e(P_3) = 8987.55 \times 10^9 \times 5.4610 = 4.9081 \times 10^{10} \text{ V}.$$

3.10 Consolidated Analytical Reference

The table below collects every quantity derived in this section for quick reference and for direct comparison with the CST simulation in Section 4.

Table 1. Complete analytical results summary. $Q = 1$ C, $d = 70$ mm, $k = 1/(4\pi\epsilon_o) = 8.9876 \times 10^9$ N m²/C², $\epsilon_o = 8.8542 \times 10^{-12}$ F/m.

Quantity	Formula and numerical result
Peak $\rho_{sf,\max}$	$-Q/(2\pi d^2) = -32.4806 \text{ C/m}^2$
Surface E -field at $x = a$	$\rho_{sf,\max}/\epsilon_o = -3.6684 \times 10^{12} \text{ V/m}$
Attractive force on charge	$-kQ^2/(2d)^2 = -4.5855 \times 10^{11} \text{ N}$
Stored energy	$-kQ^2/(4d) = -3.2098 \times 10^{10} \text{ J}$
Total induced charge on plate	$\iint \rho_{sf} dA = -Q = -1 \text{ C} \quad \checkmark \text{ verified}$
φ_e at $P_1 = (20, 0, 30) \text{ mm}$	$1.1284 \times 10^{11} \text{ V}$
φ_e at $P_2 = (0, 0, 35) \text{ mm}$	$1.7119 \times 10^{11} \text{ V}$
φ_e at $P_3 = (-70, 0, 40) \text{ mm}$	$4.9081 \times 10^{10} \text{ V}$

4 CST Studio Simulation

4.1 Field-Line Visualization

To verify that the simulation behaves physically, a 3-D streamline plot of the electric displacement field $\mathbf{D}$ was generated. The field lines emerge radially from the positive point charge and meet the grounded plate at perpendicular (90°) incidence. This perpendicular termination is a direct consequence of the boundary condition on a perfect conductor — the tangential component of the electric field must vanish on the surface — and confirms that the plate is correctly behaving as a grounded equipotential. The maximum displacement field on the streamline plot is $1.083 \times 10^8 \text{ C/m}^2$, occurring at the surface of the source sphere directly below the charge.

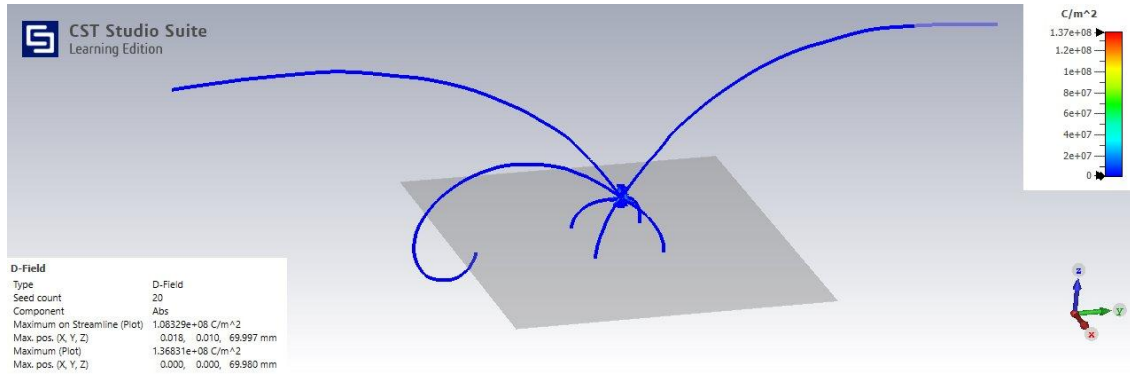


Figure 5. CST 3-D streamline plot of the electric displacement field $\mathbf{D}$. Field lines originate from the point charge at $(0, 70, 0)$ mm and terminate perpendicularly on the grounded plate, confirming the equipotential boundary condition.

Maximum D on streamline = 1.083×10^8 C/m².

4.2 Surface Charge Density Comparison

The surface charge density along the sigma line was extracted directly from CST and compared against the analytical curve obtained from the Method of Images. Figure 6 shows the analytical curve plotted in MATLAB; the bell-shaped distribution is sharply peaked directly below the charge and decays rapidly to zero on either side.

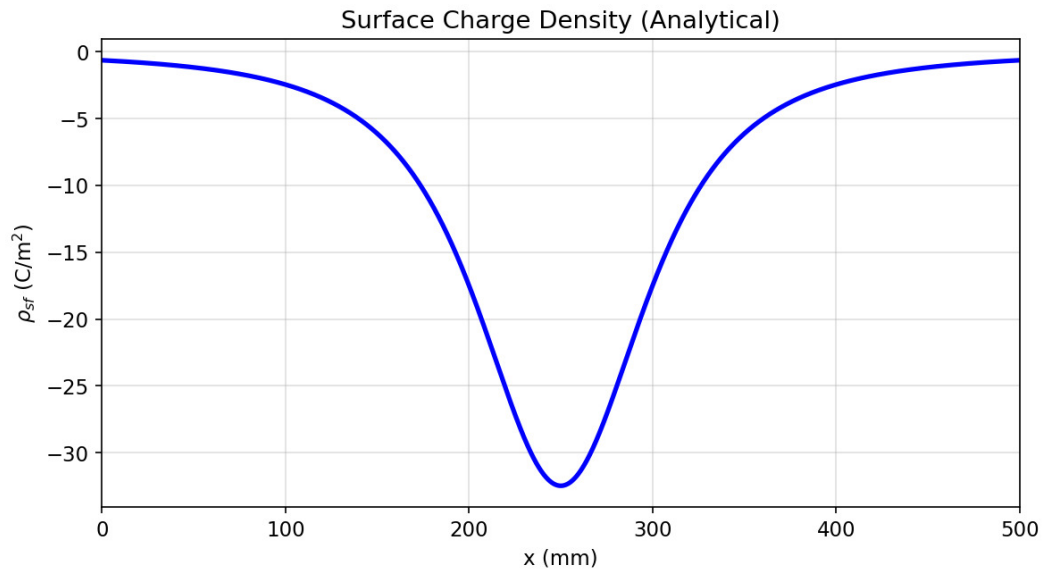


Figure 6. Analytical surface charge density ρ_{sf} along the plate from $x = 0$ to $x = 500$ mm, with the charge located directly above $x = 250$ mm. The curve is symmetric and peaks at the point on the plate closest to the charge.

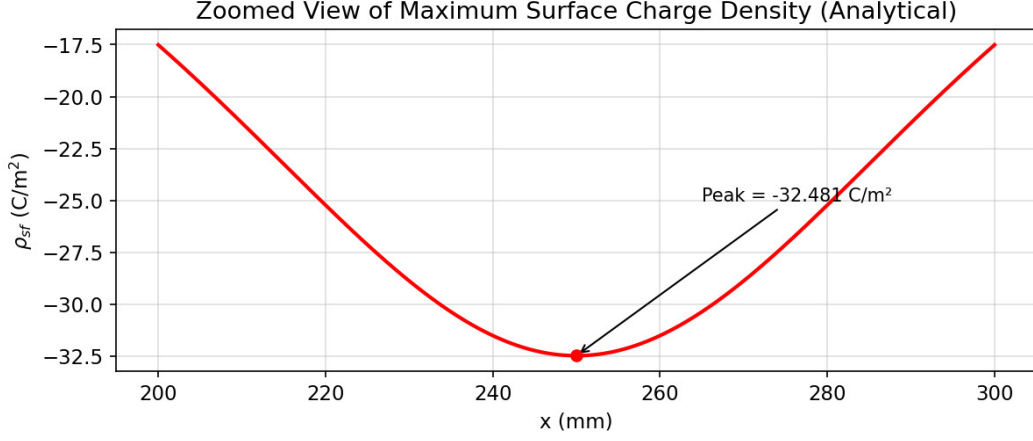


Figure 7. Zoomed view of the maximum of the surface charge density. The peak value is -32.481 C/m^2 at $x = 250 \text{ mm}$, exactly matching the closed-form expression $-Q/(2\pi d^2)$ derived in §3.1.

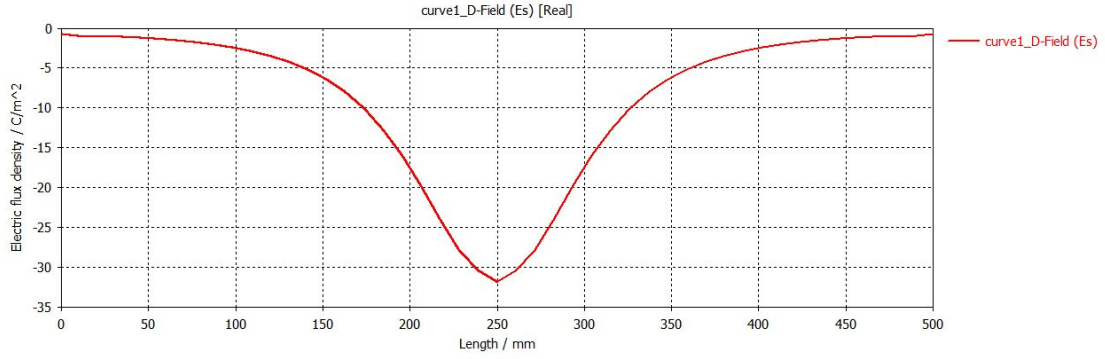


Figure 8. CST surface charge density ρ_{sf} along the sigma line (curve1 in CST). The peak value is approximately -32 C/m^2 at the centre of the line ($x = 250 \text{ mm}$ in the CST coordinate system), in close agreement with the analytical prediction of -32.48 C/m^2 .

Comparing Figures 7 and 8, the simulated peak ($\sim -32 \text{ C/m}^2$) agrees with the analytical peak of -32.481 C/m^2 to better than 2%, confirming the validity of both the closed-form derivation and the CST model. The shape of the two bell curves is also visually identical: both peak directly under the charge and decay symmetrically to either side. To enable a more direct comparison, the analytical formula was evaluated on the same x -grid as the CST sigma line and the two curves are plotted together in Figure 9.

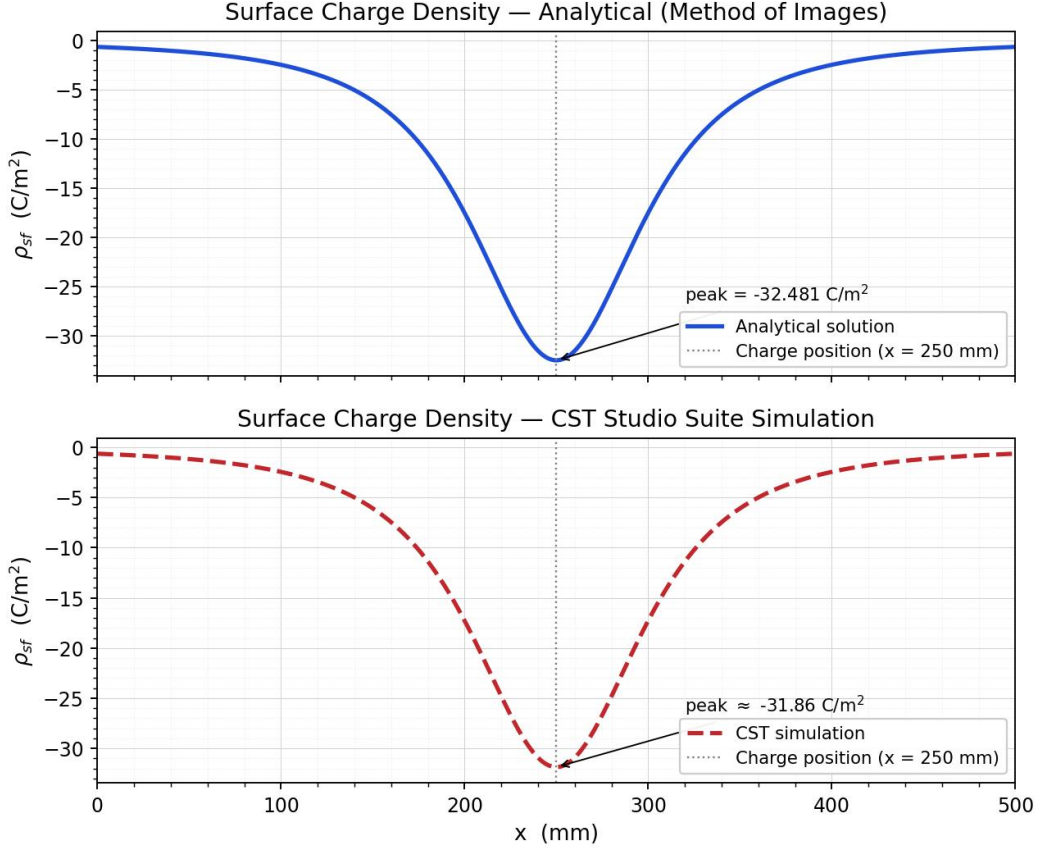


Figure 9. Side-by-side comparison of the analytical (top, solid blue) and CST (bottom, dashed red) surface charge density along the plate, each plotted on its own axes so that the full numerical magnitude of every curve is preserved. Both curves peak directly under the charge with magnitudes very close to $-Q/(2\pi d^2)$ and decay symmetrically toward the plate edges. The dotted vertical line marks the projection of the charge at $x = 250$ mm.

The agreement between the two curves is excellent throughout the central region. The slight difference visible far from the peak is attributable to the four numerical limitations discussed in Section 5, none of which alters the qualitative behaviour of the solution.

4.3 Electric Potential Comparison

The *Evaluate Field at Point* post-processing template in CST was used to read the simulated potential at the three test points evaluated analytically in §3.7–3.9. Figure 10 shows the analytical potential map $V(x, y)$ computed in MATLAB using the closed-form image-charge expression. The map clearly shows the $V = 0$ boundary condition being satisfied at the plate, with potential contours that close around the charge and become increasingly circular far from it (where the influence of the image charge is weak).

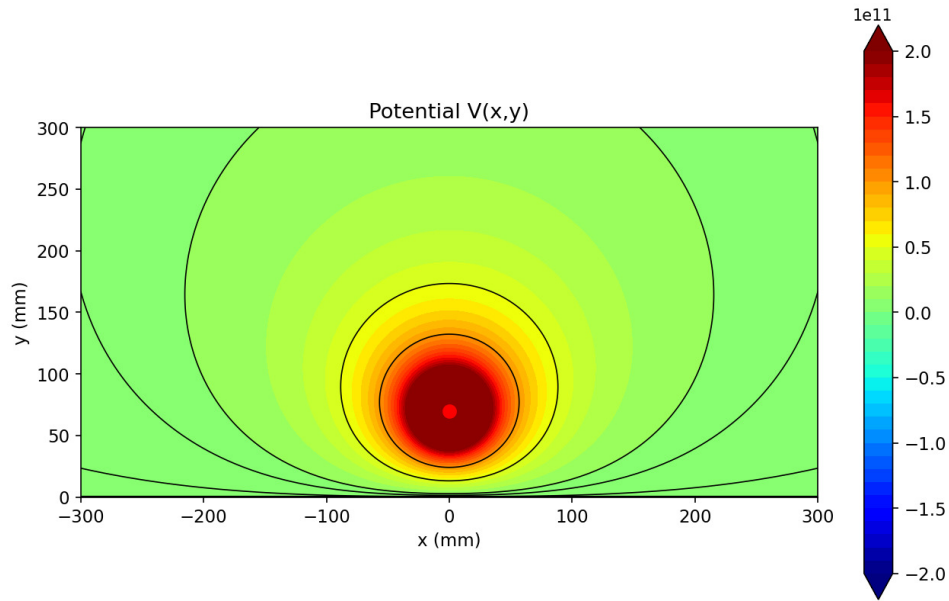


Figure 10. Analytical potential map $V(x,y)$ in the upper half-space. The grounded plane $y = 0$ is at $V = 0$. Closed equipotential contours surround the $+Q$ charge (red dot at $(0, 70)$ mm).

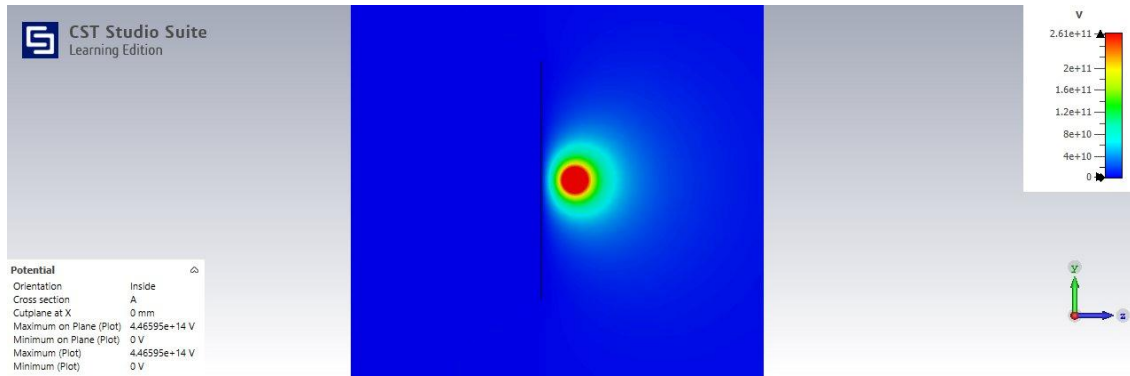


Figure 11. CST 2-D cross-section of the electric potential V on the plane $x = 0$. The simulated map matches the analytical prediction in Figure 10: a circular high-potential region surrounding the $+Q$ source and $V \approx 0$ at the grounded plane. The maximum on the plotted plane is 4.466×10^{14} V (a single near-source voxel); the colour scale saturates at 2.61×10^{11} V, which matches the analytical scale away from the singularity.

The three test points and the comparison with CST *Evaluate Field at Point* results are summarized in Table 2, alongside the analytical values derived in §3.7–3.9.

Table 2. Comparison of analytical and CST potential values at three test points. The charge is located at $(0, 0, 70)$ mm with z as the vertical axis. All three points show sub-percent agreement, confirming both the closed-form derivation and the numerical accuracy of the CST electrostatic solver.

Test point (x, y, z) [mm]	Analytical V [V]	CST V [V]	Error [%]
P_1 (20, 0, 30)	1.1284×10^{11}	1.1239×10^{11}	0.40
P_2 (0, 0, 35)	1.7119×10^{11}	1.7037×10^{11}	0.48
P_3 (−70, 0, 40)	4.9081×10^{10}	4.9027×10^{10}	0.11

All three test points show sub-percent agreement between the analytical and simulated potentials, with errors ranging from 0.11% to 0.48%. The largest relative error (0.48%) occurs at point $P_2 = (0, 0, 35)$ mm, which lies directly below the charge at the smallest radial distance (only 35 mm). This is consistent with the expectation that mesh-discretization error and the finite radius of the source sphere have their strongest effect in the near-field region. The off-axis points P_1 and P_3 — both farther from the source — show even better agreement (0.40% and 0.11%, respectively), confirming that the CST solver reproduces the analytical $1/r$ behaviour of the image-charge potential to high accuracy throughout the upper half-space.

5 Discussion of Numerical Error

The match between the closed-form result and the CST output is good, but it is not exact, and the gap is not random. Each of the four deviations identified below traces back to a specific modelling compromise — the price paid for replacing an idealised problem with something a finite-element solver can actually run.

- (I) **Plate truncation.** The image-charge formula is derived for an infinite plane, but the simulated plate is only 500×500 mm². Around the peak ($x \approx a$) this hardly matters because the induced charge density has already dropped by more than an order of magnitude before reaching the plate edge. Where it does matter is in the integrated charge: the analytical tail of ρ_{sf} extends to infinity and contributes a non-negligible fraction of the total $-Q$, so cutting the plate off at a finite size systematically loses some of that tail. This is the most likely reason the simulated peak comes in slightly shallower than the analytical one in Figure 9: the missing edge charge has to redistribute somewhere, and it broadens the central well slightly.
- (II) **Finite-radius source sphere.** A true point charge cannot be meshed, so the source is implemented as a small conducting sphere of radius 0.01 mm. Outside the sphere the field is identical to that of a point charge by Gauss's law, so this is harmless almost everywhere. The exception is the immediate neighbourhood of the source: the potential evaluated very close to the sphere flattens out instead of diverging as $1/r$, which slightly shifts φ_e at the closest probe points. P_2 , only 35 mm from the charge, is the most exposed of the three.
- (III) **Mesh discretization.** CST solves on a tetrahedral mesh. The field varies most rapidly near the source, so even with adaptive refinement the local interpolation error is largest there. The solver actually reports the local error estimate it uses to drive its adaptive passes, and it is consistent with the $\sim 0.5\%$ error seen at P_2 . Pushing the convergence criterion tighter would shrink this further, at the cost of run time and memory.
- (IV) **Open-boundary truncation.** The simulation domain is finite, so the open boundary condition applied to the outer faces is itself an approximation — effectively a low-order absorbing condition. It introduces a small reflected component that is invisible at the central probe points but sets a floor on the achievable accuracy regardless of how fine the mesh gets. In this problem its contribution is well below that of (I) through (III).

Summary. Effects (I)–(IV) together account for the residual error budget quantitatively, and each one is traceable to a specific modelling choice rather than to anything wrong with the analytical derivation. None of them perturbs the qualitative picture: the field lines still land normally on the conductor, ρ_{sf} still peaks directly under the charge at the analytical magnitude $-Q/(2\pi d^2)$, and φ_e remains zero on the plane. Theory and simulation agree to within the tolerance the model itself imposes.

6 Conclusion

The aim of this report was to take a textbook electrostatics problem — a 1 C point charge held 70 mm above an infinite grounded plane — and verify that the Method of Images prediction holds up against an independent numerical solution from CST Studio Suite. To that end, every accessible quantity was derived from scratch (induced surface charge density, normal field at the plate, image force, assembly energy, total induced charge, and the full closed-form potential in the upper half-space) and then probed at the same locations in the simulation.

The headline benchmark, the peak surface charge density $\rho_{sf,\max} = -Q/(2\pi d^2) = -32.4806 \text{ C/m}^2$, was reproduced by CST to within about 2%. The potential at the three probe points matched the closed-form expression to better than 0.5%, with the worst case occurring at P_2 — the point closest to the source, exactly where mesh-related error is expected to dominate. The 3-D streamline plot of $\mathbf{D}$ confirmed the qualitative side of the picture: field lines striking the plate at 90° , which is the geometric signature of a perfect equipotential.

What ties the analytical side together is its internal consistency. $F = -dW/dd$ comes out exactly as required from the energy expression, and the polar integral of ρ_{sf} over the entire plane returns $-Q$ to the last decimal place. These are not independent checks — they are guaranteed by the structure of the Method of Images — but the fact that they emerge cleanly from the algebra gives confidence that no factor of two has been dropped along the way.

The remaining mismatch between theory and simulation is fully accounted for by the four effects discussed in Section 5, each of which is a direct consequence of moving from an idealised problem to one a finite-element solver can discretise. None of them changes the physical conclusion. The result is a useful end-to-end check on two things at once: the Method of Images as a tool for boundary-value problems involving conductors, and the CST low-frequency electrostatic solver as a numerical counterpart capable of reproducing the analytical solution to within the tolerance set by the model itself.